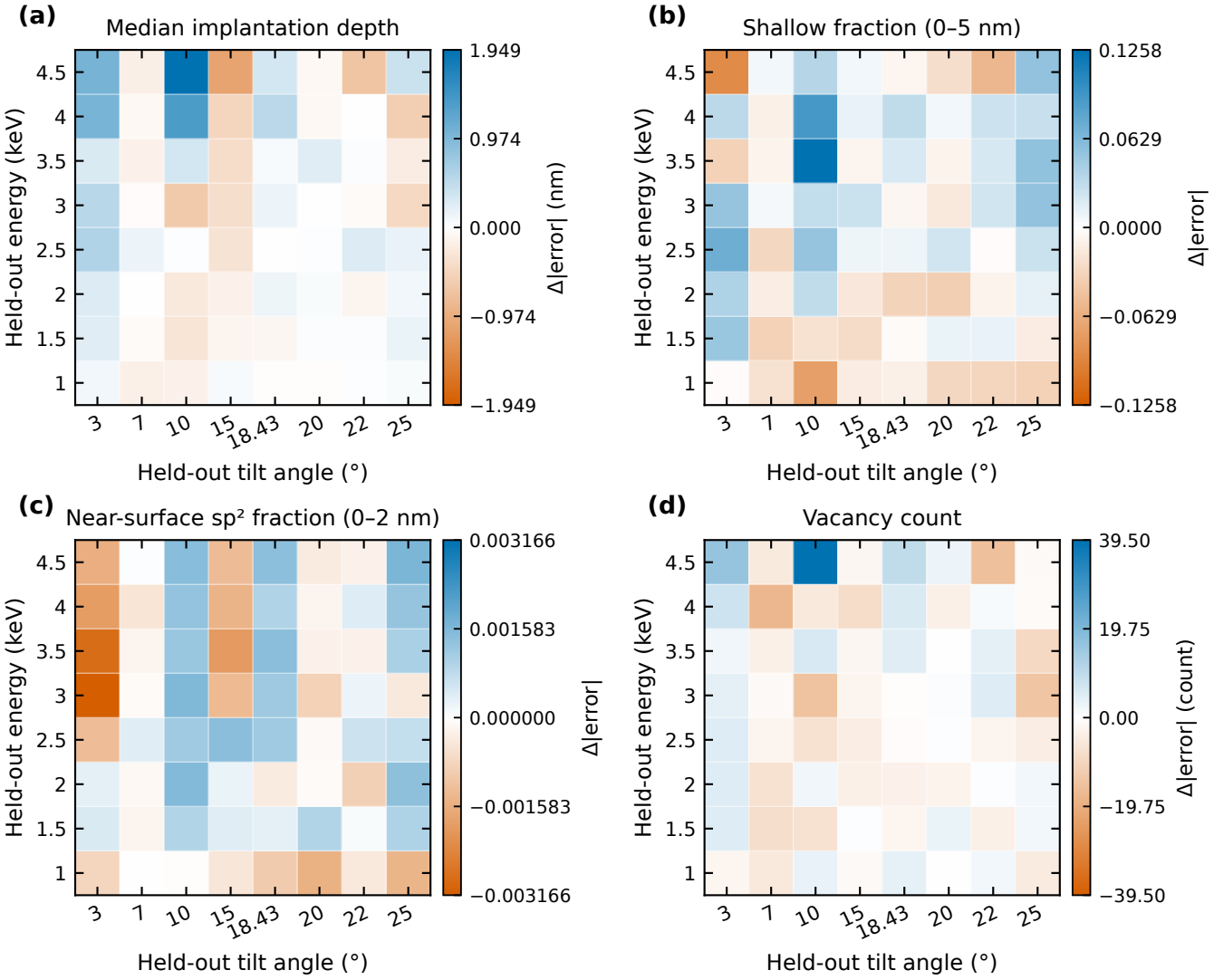

(a)
Median implantation depth
Held-out energy (keV)
4.5
4
3.5
3
2.5
2
1.5
1
3
1
10
15
18.43
20
22
25
Held-out tilt angle (°)
1.949
0.974
0.000
−0.974
−1.949
Δ|error| (nm)
(b)
Shallow fraction (0–5 nm)
Held-out energy (keV)
Held-out tilt angle (°)
0.1258
0.0629
0.0000
−0.0629
−0.1258
Δ|error|
(c)
Near-surface sp² fraction (0–2 nm)
Held-out energy (keV)
Held-out tilt angle (°)
0.003166
0.001583
0.000000
−0.001583
−0.003166
Δ|error|
(d)
Vacancy count
Held-out energy (keV)
Held-out tilt angle (°)
39.50
19.75
0.00
−19.75
−39.50
Δ|error| (count)